Superstore Data Analysis (Task 1)

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Data Analysis

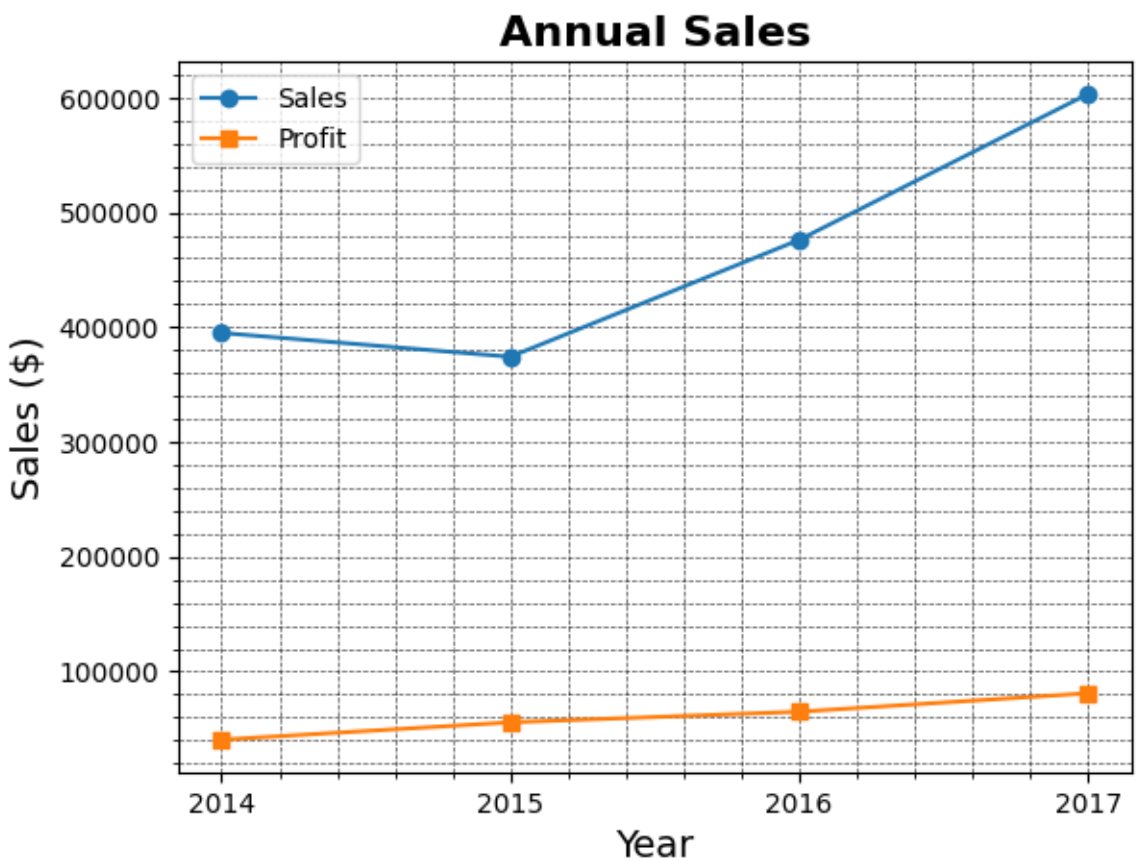
Preview

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell
6	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman

Revenue

This section analyses the sales and profit made throughout the year and aims to give clear insights on the periods at which the business does well and the periods at which it does not.

Annual Sales



Key Observations

- Anual Sales

  Annual sales show a clear upward trend from 2015 to 2017, indicating strong business growth after a slight dip in 2015.

  The significant increase between 2016 and 2017 suggests that business strategies implemented during this period were highly effective in driving revenue.

- Profit

  Profit follows a consistent upward trend, increasing each year.

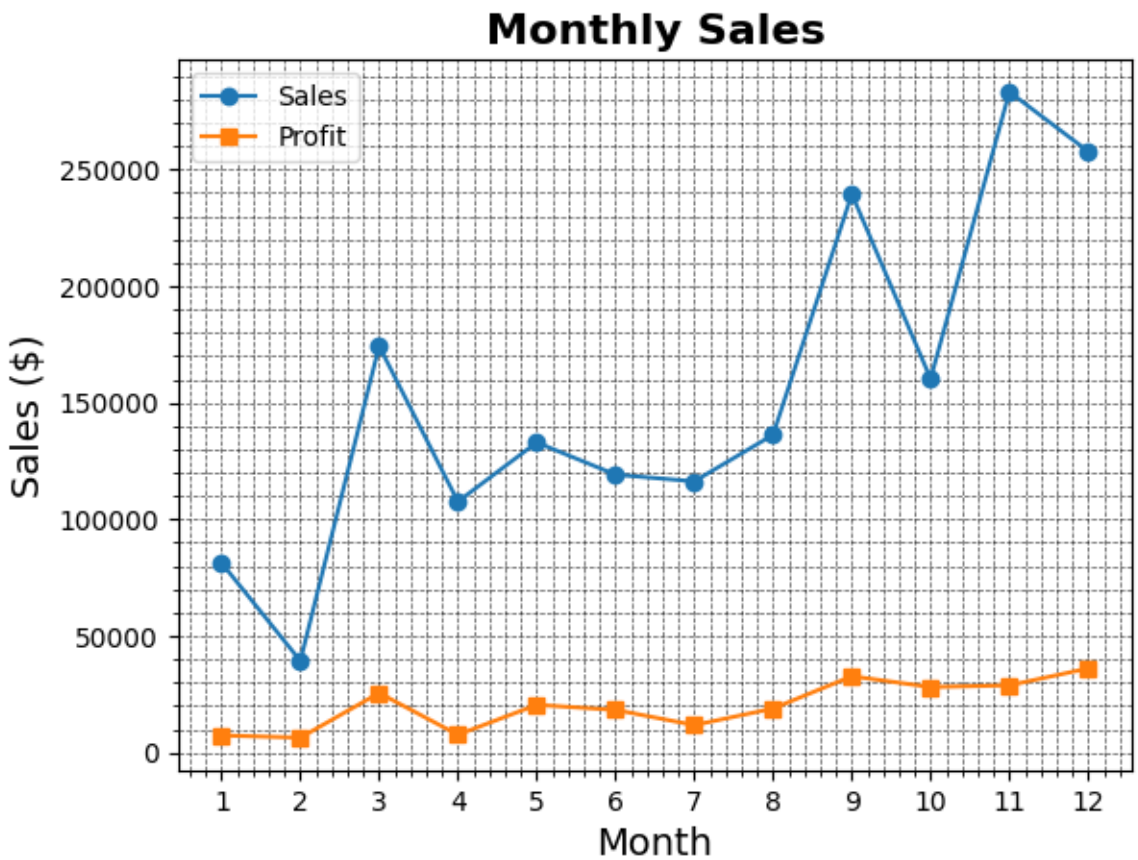
  This indicates that the business is not only generating more revenue but is also maintaining profitability as it scale

  Although, there has been a significant gap between the anual sales measured to be approximately, 345k in 2014 and 522k in 2017, meaning the gap has increased

Insight

Overall the anual sales grow at a greater rate than anual profit. This could be due to a number of factors, such as increased costs, lower profit margins, or a shift towards lower-margin products. Further analysis is required to identify the exact cause and improve overall profitability.

Monthly Sales



Key Observations

- Early-Year Dip

  Both sales and profit are significantly lower in January and February, with February being the lowest-performing month.

  This suggests a post-holiday slowdown, where customer spending typically decreases after the festive season.

- Mid-Year Recovery

  From March onward, there is a sharp increase in sales, followed by moderate fluctuations between April and August.

  This indicates a period of stabilized business activity, where sales are consistent but not at peak levels.

- Late-Year Surge

  A strong upward trend is observed from September through December, with:

  1. A major peak in November
  2. Sustained high performance in December

  Profit follows a similar pattern, confirming that this is the most valuable period for the business.

Insight

The business shows clear seasonal behavior:

1. Weak performance at the start of the year
2. Strong performance toward the end of the year

Reccomendations

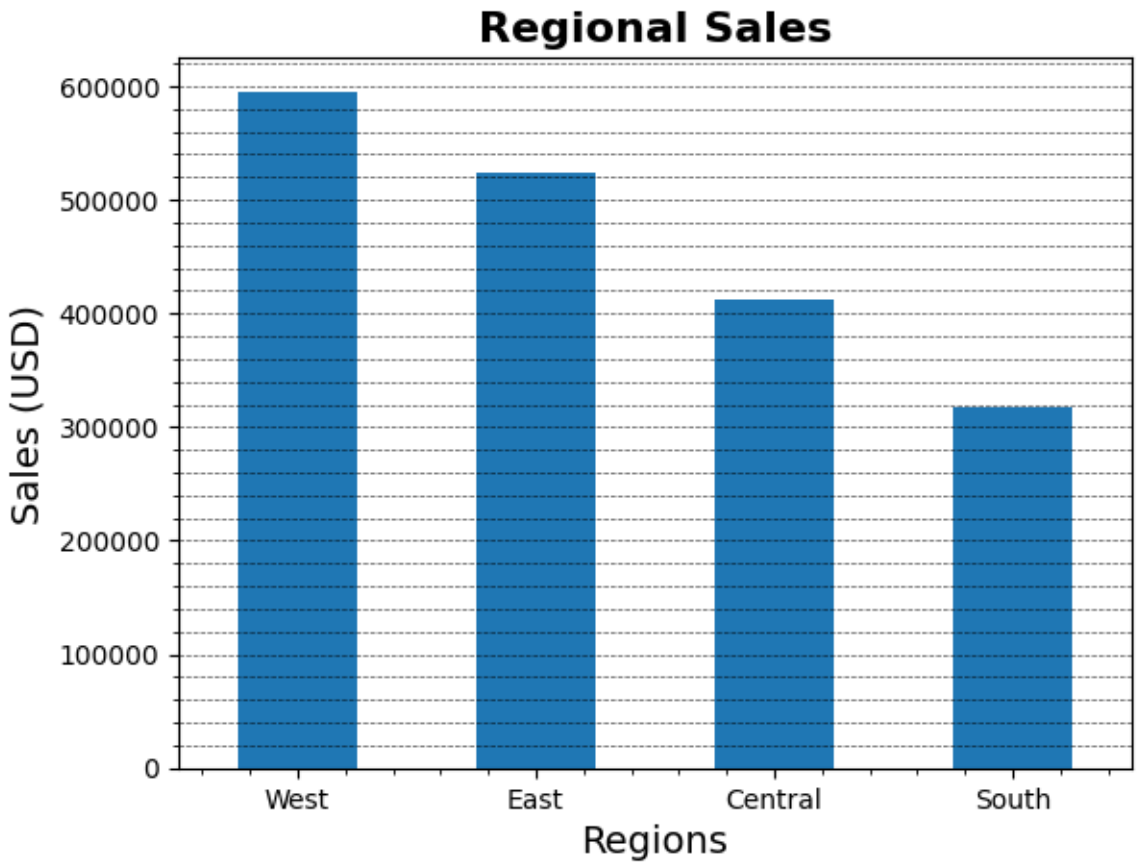
To improve overall performance, the business should **focus on boosting sales during the slower early months** while fully capitalizing on the high-demand period at the end of the year.

Regional Perfomance

Overview

This section analyzes sales performance across 4 regions: West, East, Central, and South using Sales, Profit, and Profit Margin.

Sales by Region



Key Observations

- Top Performing Region

  The West region generates the highest sales, followed closely by the East.

  This indicates that these regions are the **primary drivers** of revenue for the business.

- Mid-Level Performance

  The Central region shows moderate performance, contributing a significant portion but still trailing behind the top regions.

- Underperforming Region

  The South region records the lowest sales, generating substantially less revenue than all other regions.

  Compared to the West, the South generates approximately 47% less revenue, highlighting a clear performance gap.

Insight

There is a clear imbalance in regional performance, with revenue concentrated in the West and East.

This could be due to:

1. Differences in market demand
2. Unequal resource allocation
3. Variations in marketing effectiveness
4. Regional economic factors

Recommendations

1. Investigate South Region Performance

   1. Analyze customer behavior and demand in the South
   2. Evaluate whether the region is under-marketed or under-supplied
   3. Identify barriers such as pricing, logistics, or product fit

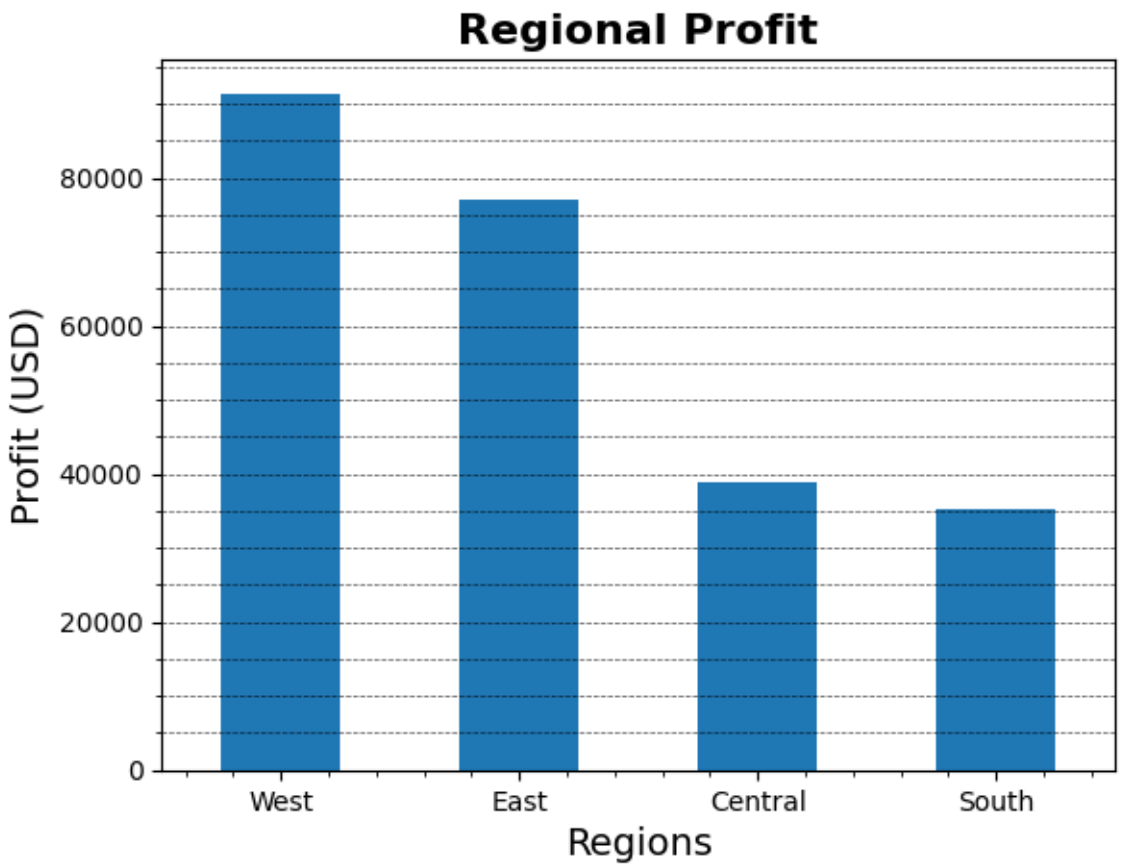
2. Leverage High-Performing Regions

   Study strategies used in the West and East and consider implimenting them in the under performingregions

3. Optimize Resource Allocation

   Consider redistributing marketing and operational efforts Focus on high-growth potential regions while improving weaker ones

Profits by Regions



Key Observations

- Top Performing Regions

  The West region generates the highest profit, approximately $91k, followed by the East, consistent with the trend observed in sales.

  This indicates that these regions are not only generating high revenue but are also maintaining strong profitability.

- Central Region Concern

  The Central region generates relatively high sales (approx. 412k) but significantly lower profit (approx. 8.9k).

  This suggests a low return on sales, indicating potential issues such as:

  1. High operational or shipping costs

2. Heavy discounting
3. Lower-margin products

- South Region Efficiency

  Although the South region has the lowest sales (approx. 317k), its profit (approx. 35k) is very close to that of the Central region.

  This indicates that the South is operating with better efficiency, achieving similar profit levels with significantly lower sales.

Insight:

There is a clear difference between revenue generation and profit efficiency across regions.

The Central region appears to be underperforming in terms of profitability The South region, despite lower sales, is relatively more efficient

Recommendations:

1. Investigate Central Region Costs

   Analyze cost structure (logistics, discounts, operations) Identify areas where profit margins are being reduced Optimize pricing or reduce unnecessary expenses

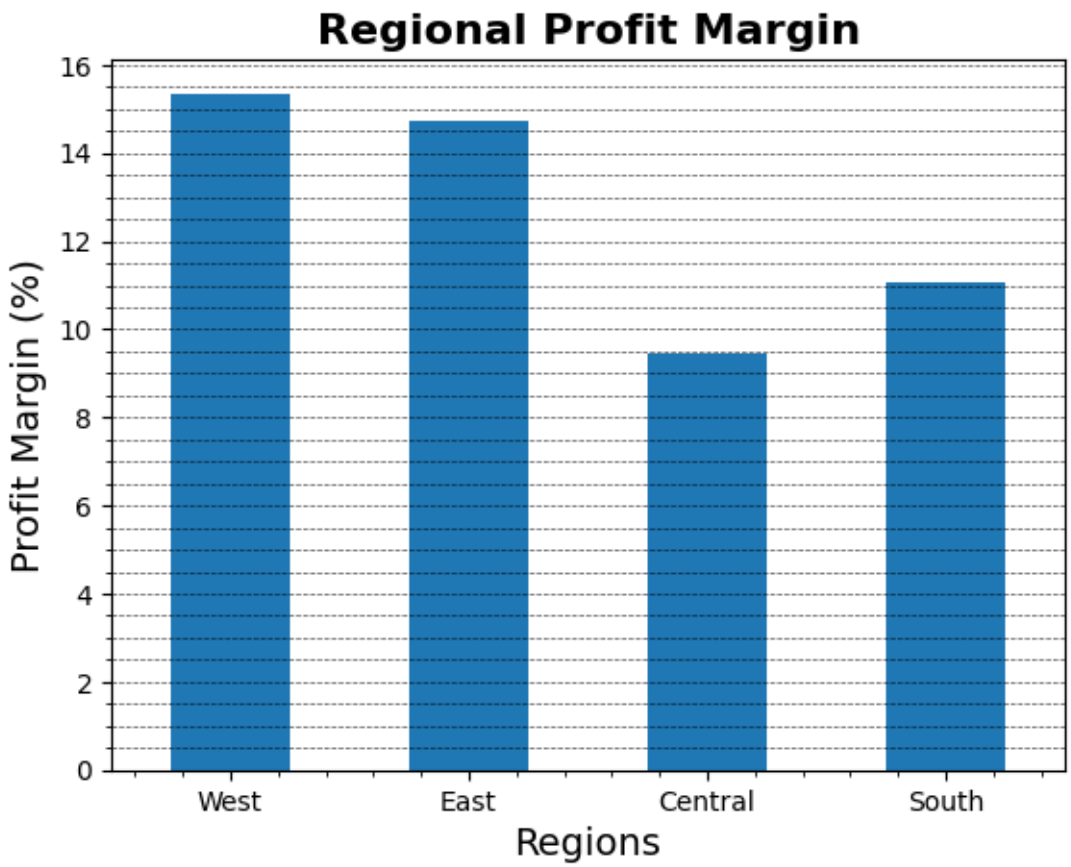
2. Learn from South Region Efficiency

   Identify what is working well in the South Replicate efficient practices in other regions, especially Central

3. Strengthen High-Profit Regions

   Continue investing in West and East, as they drive both revenue and profit Scale successful strategies from these regions

Profit margins



Key Observations

- High-Performing Regions

  The West and East regions have the highest profit margins, indicating strong cost control and efficient operations.

- Moderate Performance

  The South region shows moderate efficiency, maintaining a reasonable margin despite lower overall sales.

- Critical Concern: Central Region

  The Central region has the lowest profit margin (9.44%), confirming earlier observations that it is underperforming in terms of profitability.

  This reinforces that:

1. High sales high profit
2. The Central region is likely experiencing cost inefficiencies or aggressive discounting

Insight

Profit margin reveals the true performance of each region:

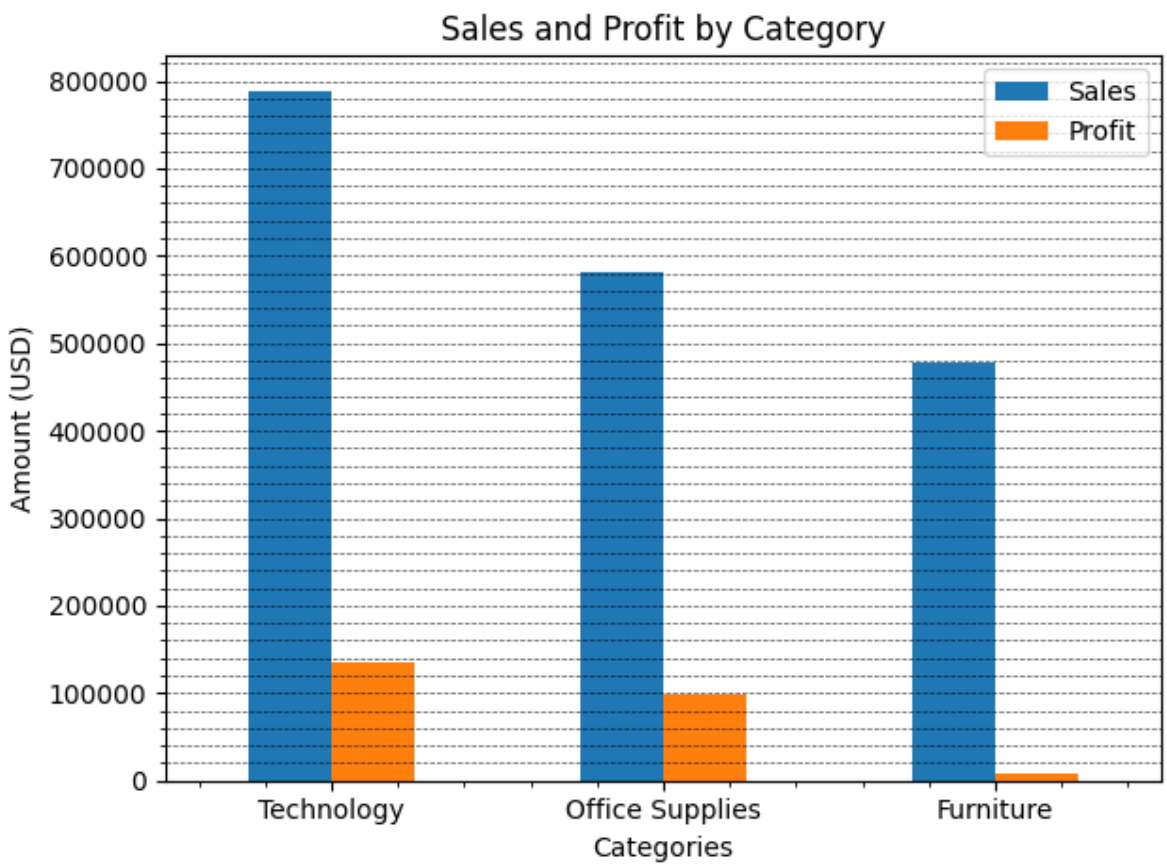
West/East: Strong revenue and strong efficiency which is why they are best-performing regions

South: Lower revenue but decent efficiency so, it is fairly stable Central: High revenue but poor efficiency Meaning there is prioritising issues

Recommendations

1. Fix Central Region Profitability (High Priority)
    1. Review products in the Central region with:
        1. Discount rates > 15%
        2. Low profit margins
    2. Target: Increase Central region profit margin from 9.44% → at least 12% within 2 quarters
2. Audit pricing strategies in Central:
    1. Identify products consistently sold at reduced margins
    2. Implement minimum margin thresholds to prevent loss-leading sales
3. Shift focus toward:
    1. High-margin products
    2. Bundled offerings instead of heavy discounts
    3. Reducig reliance on low-margin, high-volume items
4. Replicate High-Performing Region Strategies

Categories



Key Observations

The ratio between the sales vs profit is large due to category being broad.

- Top Performing Category: Technology

  Technology generates the highest sales and the highest profit by a large margin.

  This indicates:

  1. Strong demand
  2. High-margin products
  3. Effective pricing strategy

- Stable Performer: Office Supplies

  Office Supplies shows solid performance in both sales and profit, making it a reliable contributor to overall business revenue.

- Critical Issue: Furniture Category

  Furniture generates significant sales (~478k) but extremely low profit (approx. 8.9k).

  This a profit margin of roughly:

  ~1.9% (very low compared to other categories)

  This suggests:

  1. High costs (shipping, production, storage)
  2. Heavy discounting
  3. Low-margin products

Insight

Technology contributes the most to business growth and is the profit driver. Office Supplies are the stable backbone pf the business. Lastly Furniture is where the problem area (revenue without value)

Recommendations

1. Fix Furniture Profitability (High Priority) Identify top-selling furniture products with low margins
2. Reduce Loss-Making or Low-Margin Products
3. Increase marketing and inventory for Technology products Expand product range in high-performing sub-categories

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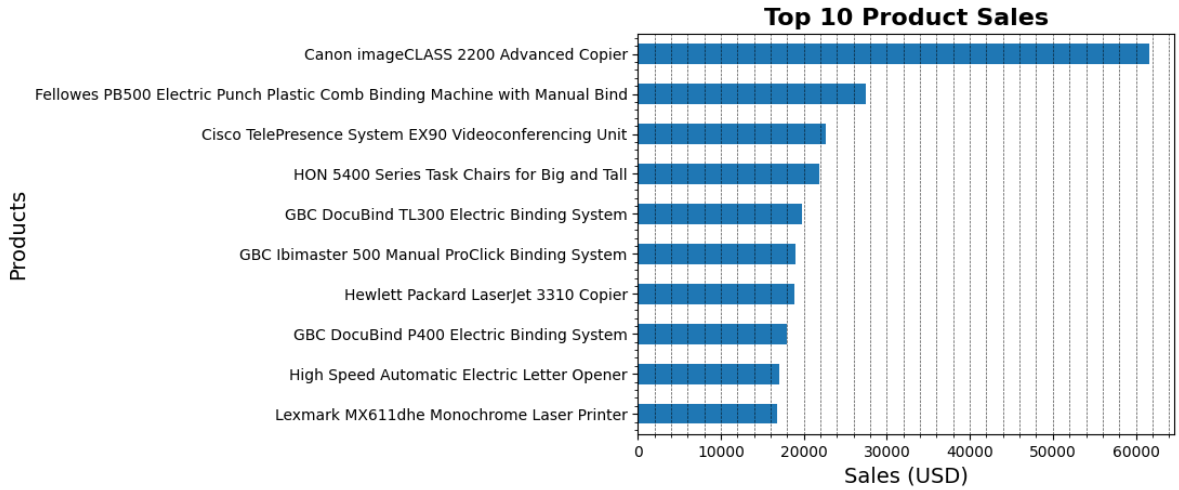
Top-Selling Products



Key Observations

- Technology Dominance

  Many of the top products fall under the Technology category, reinforcing earlier findings that technology is the **primary revenue driver** of the business

- Furniture Presence

  Products like task chairs appear among top sellers, but earlier analysis showed Furniture has low profit.

  This suggests furniture products may sell well, but are likely low-margin or high-cost

Insight

High-value products seem to drive revenue A strong Technology segment for profitability

However, some top-selling products may not be highly profitable, especially in categories like Furniture

Recommendations

1. Identify which of these top-selling products: Have low or negative profit margins
2. Prioritize High-Profit Products Focus marketing and promotions on: Products with both high sales AND high profit
3. Optimize supply chain costs
4. Bundle high-margin + high-demand products

5. Invest more in Technology Category since its products are the top sellers

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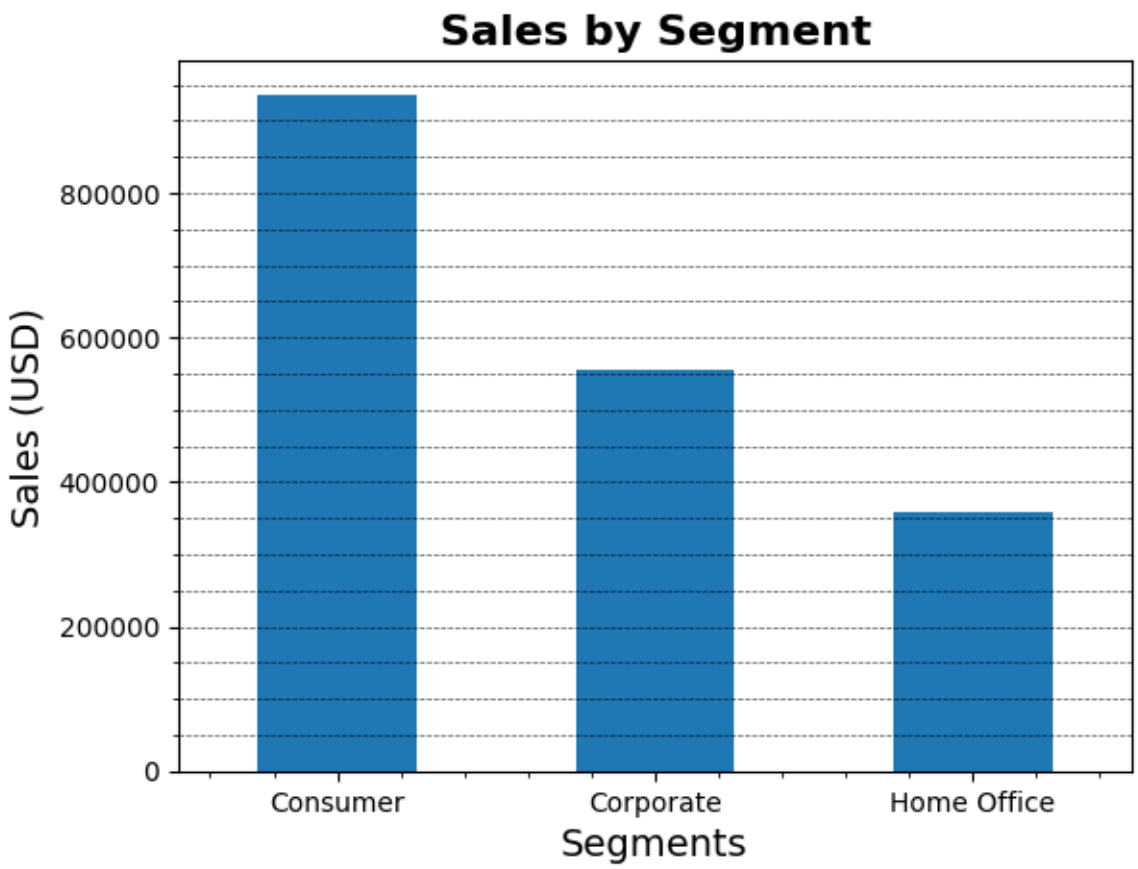
Segments

Sales



Key Observation

- Primary Revenue Driver

  The Consumer segment generates the highest sales compaared to the Corprate and Home Office segments.

  This indicates that individual customers are the main source of revenue for the business.

- Secondary Contributor

  The Corporate segment contributes a substantial portion of sales, but still trails behind Consumer customers.

  This suggests steady demand from businesses, though not at the same scale as individual buyers.

- Lowest Performing Segment

  The Home Office segment generates the lowest sales.

  This may indicate lower demand or less targeted marketing toward this segment

Insight

The business relies heavily zon the Consumer segment, making it the core revenue engine.

However, this also introduces risk such as over-dependence on one segment.

Recommendations

1. Continue investing in Consumer Segment (Maintain Growth)

   - Marketing campaigns
   - Promotions
   - Product availability

   Since this is your largest revenue driver — protect and grow it

2. Expand Corporate Segment Opportunities by introducing long-term contracts.

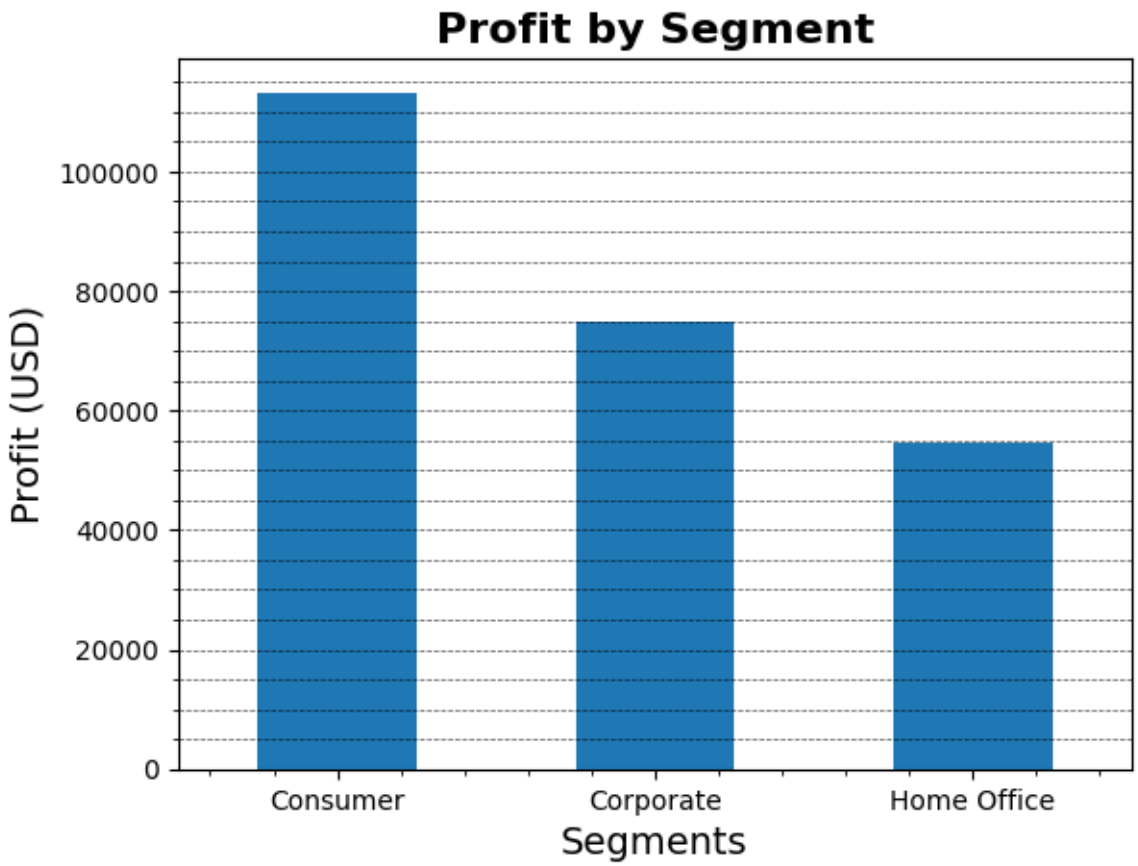
3. Grow Home Office Segment by developing targeted campaigns for:

   1. Remote workers
   2. Small/home-based businesses
   3. Offer bundled office setups or affordable product packages

4. Segment-Specific Strategy

   - Avoid a using one approach for every segment. Review the statistics of the perfomance of each segment and strategise accordingly.
   - Tailor pricing, promotions, and products for each segment

Profit



key Observations

- Consistency

  The profits remain consistent with the sale. Although the Home Office segment has the lowest sales, it still generates a notable level of profit (approximately 54k) suggesting potentially higher margins compared to its size

Insight

When comparing sales vs profit across segments:

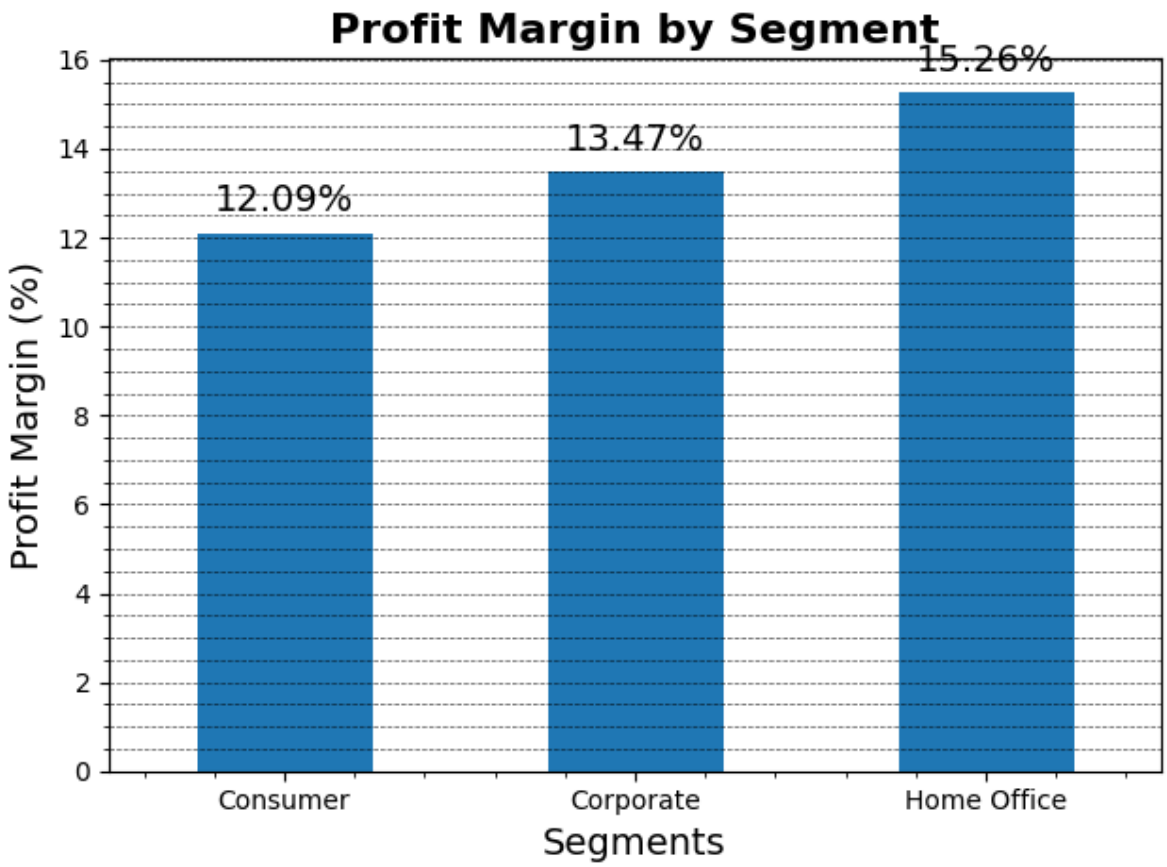
Consumer: - Have a high sales and high profit making them the **core growth engine** Corporate: - Have moderate sales and profit making Corporate the **stable contributor** Home Office: - Have the lowest sales but decent profit making them have **unnoticed potential**

This tells that the business is not just driven by volume (Consumer), but also has efficient smaller segments (Home Office) that can be expanded.

Recommendations

1. Continue Scaling Consumer Segment by maintaining strong marketing and product availability
2. Strengthen Corporate Relatiionships by utilising the long term contract metioned previously
3. **Expand Home Office Segment:**
   - Target, remote workers, freelancers, small businesses, etc and capitalize on them
   - Offer affordable starter kits to help attract clients of the Home Office Segment, which has the highest potential
   - Increase sales while maintaining current efficiency
   - Since technology has the highest sales, the business should consider investing more of it into this segment
4. Segment Profitability Strategy -Analyze profit margins per segment; which segment converts revenue into profit most efficiently

Profit Margin



KPIs used

The following key performance indicators (KPIs) were used to evaluate business performance:

Total Sales

Measures overall revenue generated across different dimensions (time, region, category, and segment).

Total Profit

Indicates the net earnings after costs, used to assess overall profitability. Profit Margin (%)

Calculated as (Profit / Sales × 100), this metric evaluates how efficiently the business converts revenue into profit.

Sales Growth (Annual Trend)

Tracks changes in revenue over time to identify business growth patterns. Seasonal Trends (Monthly Sales & Profit) Identifies fluctuations in performance throughout the year.

Regional Performance

Compares sales and profit across regions to detect geographical strengths and weaknesses. Category & Product Performance Evaluates which product categories and individual products drive revenue and profit.

Customer Segment Performance

Analyzes contribution from different customer groups (Consumer, Corporate, Home Office). Compared the rates between the sales an profits of each segment by calculating: (each segment profit / each segment sale) and ploting that

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Key Takeaways

The analysis reveals that the business is experiencing **steady growth in both sales and profit**, although profit is increasing at a slower rate, indicating a **decline in efficiency**. There is a clear **seasonal pattern**, with performance peaking toward the end of the year and dipping in the early months. Regionally, **the West and East drive the majority of revenue**, while the **Central region underperforms in profitability**, highlighting inefficiencies. From a product perspective, **Technology is the strongest category**, whereas **Furniture generates high sales but very low profit**, making it a key concern. The **Consumer segment is the primary revenue and profit driver**, though opportunities exist to expand the **Home Office segment** and leverage the **efficiency observed in the South region**. Overall, the business should **focus on improving low-margin areas while scaling its high-performing segments and categories**.